

# Linux Filesystem

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## What is Linux Filesystem

The Linux Filesystem is a hierarchical structure starting at the root directory (/). Everything in the system—files, directories, devices, and processes—exists under this unified structure. Unlike other operating systems that use separate drive letters, Linux organizes all resources (including external devices) under a single directory tree.

In Linux, everything is treated as a file, including hardware devices, which simplifies interactions. Directories like /bin, /home, and /etc organize system binaries, user files, and configuration settings respectively, making the system flexible and consistent to manage.

## Key Directories in the Linux Filesystem:

1. / (Root Directory): The top of the filesystem hierarchy. All other directories and files are located under it.
2. /bin (Binary Directory): Contains essential command binaries (executable programs) required for system operation. Commands like ls, cp, mv, etc., are found here.
3. /boot: Contains files required to boot the system, including the kernel, bootloader (e.g., GRUB), and configuration files.
4. /dev (Device Files): Contains device files, which represent hardware components such as hard drives, terminals, and USB devices. Linux treats hardware as files, allowing programs to interact with them via I/O operations.
5. /etc (Configuration Files): Contains system-wide configuration files and shell scripts used for startup and system management. Files like passwd, fstab, and directories like cron are located here.
6. /home: Contains personal directories for all users on the system. For example, /home/user1 would be the home directory of user1. Users' personal files, configurations, and data are stored here.
7. /lib (Library Files): Contains shared libraries and kernel modules required by the essential binaries in /bin and /sbin. These are comparable to DLL files in Windows.
8. /Media: Temporary mount points for removable media such as USB drives, CDs, and DVDs.
9. /mnt: A generic mount point used for mounting temporary filesystems like network shares or partitions.

10. `/opt` (Optional Software): Contains add-on software packages. Typically used for third-party applications that aren't part of the default distribution.
11. `/proc` (Process Information): A virtual filesystem that provides real-time system information about processes. Files here represent system and process attributes like CPU, memory, and system state.
12. `/root`: The home directory of the root user (superuser). It's separate from `/home` to ensure that the system administrator's files are isolated.
13. `/run`: Contains data that describes the system since it was booted. This includes process IDs (PIDs), system information, and service status.
14. `/sbin` (System Binaries): Contains essential system binaries and administrative commands that are mostly used by the superuser. Examples include `reboot`, `shutdown`, and `fsck`.
15. `/srv` (Service Data): Contains data served by the system, often used by web servers or file servers to store website data or shared files.
16. `/sys`: Another virtual filesystem, similar to `/proc`, providing information about the system and its hardware. Used by system utilities to monitor and configure devices and components.
17. `/tmp` (Temporary Files): A temporary directory where programs store files that are needed only temporarily. This directory is usually cleared upon system reboot.
18. `/usr` (User Binaries and Programs): Contains user-installed software and files, including binaries, libraries, documentation, and source code for applications. Subdirectories include `/usr/bin` (user binaries), `/usr/lib` (user libraries), `/usr/share` (shared files), etc.
19. `/var` (Variable Files): Stores variable data that changes frequently during system operation, such as log files (`/var/log`), caches, spool files (like printer queues), and temporary email files.

## Common Linux Terminal Commands

1. `ls` (List): Lists the contents of a directory. Options like `ls -l` provide detailed information, including file permissions, size, and modification date.
2. `cd` (Change Directory): Used to change the current working directory. For example, `cd /home/user` navigates to the user's home directory.
3. `pwd` (Print Working Directory): Displays the current working directory.
4. `cp` (Copy): Copies files or directories. Example: `cp file1 /destination/directory`.
5. `mv` (Move or Rename): Moves or renames files or directories. Example: `mv file1 newname` renames `file1` to `newname`.

6. **rm (Remove):** Deletes files or directories. The `-r` option allows for recursive deletion of directories. Example: `rm -r directory`.
7. **mkdir (Make Directory):** Creates new directories. Example: `mkdir new_directory`.
8. **rmdir (Remove Directory):** Removes empty directories.
9. **cat (Concatenate and Display):** Displays the contents of a file. Example: `cat file.txt` shows the file contents in the terminal.
10. **more/less:** Paginates the output of long files. `more` scrolls forward, and `less` allows scrolling up and down.
11. **Echo:** Displays a line of text. Often used to print variables or strings in the terminal. Example: `echo "Hello, World!"`.
12. **Find:** Searches for files and directories within a directory hierarchy. Example: `find /home -name "*.txt"` finds all `.txt` files under `/home`.
13. **Grep:** Searches for a specified string or pattern within files. Example: `grep "error" logfile.txt`.
14. **ps (Process Status):** Displays information about active processes. Example: `ps aux` shows detailed information about all processes running.
15. **Top:** Provides a dynamic, real-time view of system processes, showing resource usage.
16. **chmod (Change Mode):** Changes file permissions. Example: `chmod 755 file.sh` grants the owner read/write/execute permissions and others read/execute.
17. **chown (Change Owner):** Changes file ownership. Example: `chown user:group file`.
18. **df (Disk Free):** Displays disk space usage. Example: `df -h` shows the usage in a human-readable format (e.g., GB or MB).
19. **du (Disk Usage):** Displays the size of directories and files. Example: `du -sh /directory` shows the total size of the directory in a human-readable format.
20. **tar (Tape Archive):** Used to create, extract, and manipulate archive files. Example: `tar -czvf archive.tar.gz /directory` creates a compressed archive of the directory.
21. **sudo (Superuser Do):** Executes commands as another user (often as the superuser). Example: `sudo apt-get update`.
22. **apt/yum/dnf:** Package management commands for installing, updating, and removing software. Example: `sudo apt install package_name`.
23. **kill:** Sends signals to terminate or control processes. Example: `kill 1234` terminates the process with PID 1234.

24. `man` (Manual): Displays the manual for a command. Example: `man ls` shows the manual page for the `ls` command.
25. `ssh`: Connects to a remote server via SSH (Secure Shell). Example: `ssh user@server.com`.